$$\cdot \; \varepsilon \dot{w} = -u_2 + u_3$$

$$\dot{w} = \frac{-u_2 + u_3}{\varepsilon}$$

$$= \left(c_3 \frac{w}{x} - c_4 x\right) + \frac{\left(C_{ext}(t) - w\right) - c_6 L(t) w + c_5 L_k \frac{w\Gamma}{w + w\Gamma}}{\exp(\alpha w)}$$

$$D_{u_2} \rightarrow q_x = \frac{-c_3 \frac{w}{x^2} - c_4}{\varepsilon}$$

$$q_w = \frac{c_3}{\varepsilon x}$$

$$D_{u_3} \rightarrow f_w = \frac{\frac{-w\alpha}{\exp(\alpha w)^2} - \frac{1}{\exp(\alpha w)} + \frac{2\Gamma^2 w}{w^2 + 2w^2\Gamma + w^2\Gamma^2}}{\varepsilon}$$

$$\frac{-w}{\exp(\alpha w)}$$

$$-w \cdot \frac{1}{\exp(\alpha w)}$$

$$-w \frac{\partial}{\partial w}\left(\frac{1}{\exp(\alpha w)}\right) + \frac{\partial}{\partial w}(-w)\left(\frac{1}{\exp(\alpha w)}\right)$$

$$\frac{w + w\Gamma \frac{\partial}{\partial w}(w\Gamma) - w\Gamma \frac{\partial}{\partial w}(w + w\Gamma) c_6 L(t)}{(w + w\Gamma)^2}$$

$$\frac{-w\alpha}{\exp(\alpha w)^2} - \frac{1}{\exp(\alpha w)}$$

$$\frac{\Gamma w + \Gamma^2 w - w\Gamma + \Gamma^2 w}{w^2 + 2w^2\Gamma + w^2\Gamma^2} = \frac{2\Gamma^2 w}{w^2 + 2w^2\Gamma + w^2\Gamma^2}$$

Da.1

$f_x = C_1$

$f_y = f'(y)T(t)y + T(t)f(y)$

$- c_2 \frac{\partial}{\partial y}\left[f(y)\,T(t)\,y\right]$

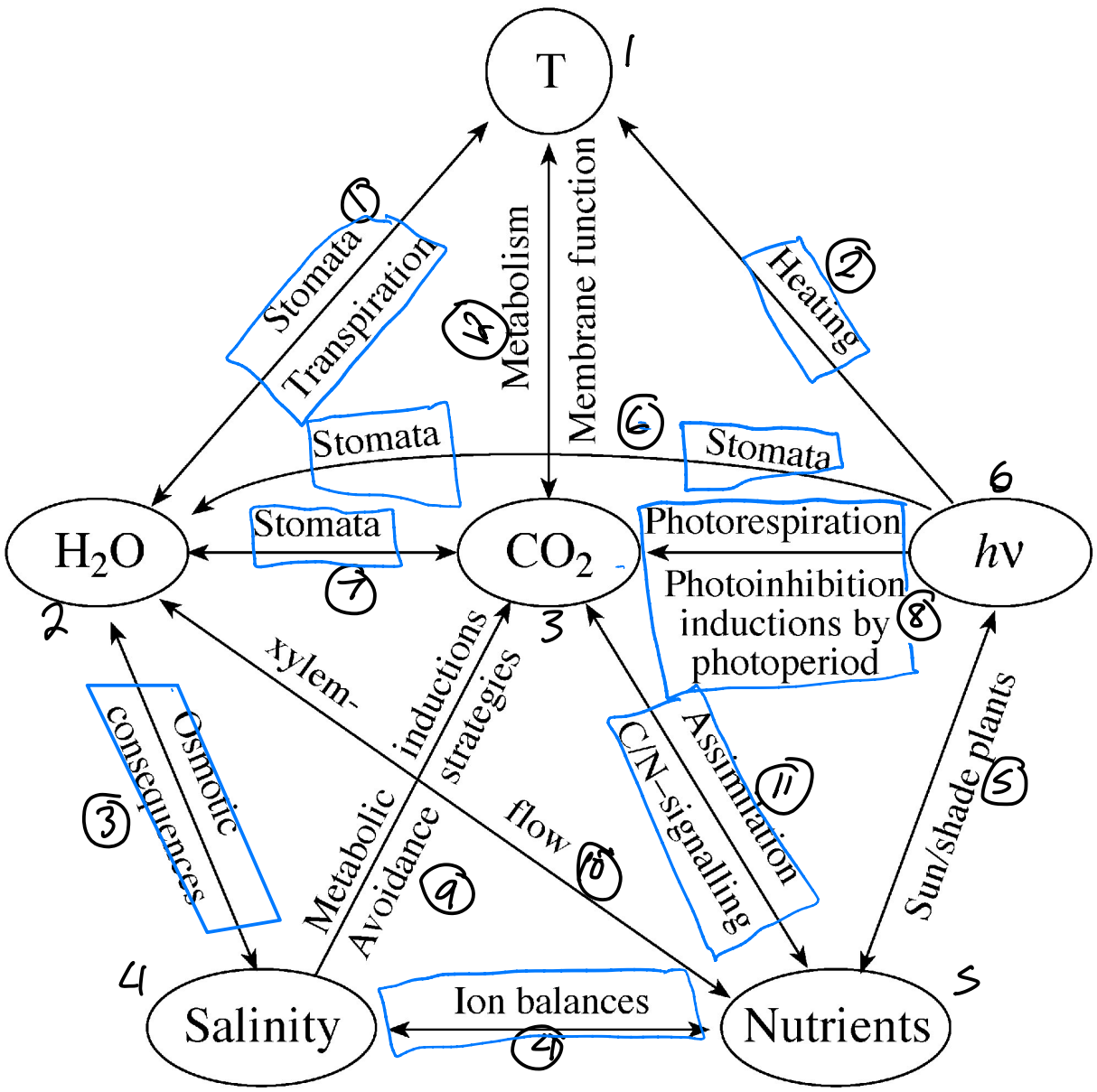
$\frac{\partial}{\partial y} f(y) \cdot T(t)y + \frac{\partial}{\partial y} T(t)y \cdot f(y)$

$f'(y)T(t) + T(t)f(y)$

Gradients

- $\left(\frac{-\frac{C_3 w}{x^2} - C_4}{\epsilon}\right)\hat{i} + \frac{C_3}{\epsilon x}\hat{j} - \left(\frac{w_2}{\exp(dw)^2}\frac{-1}{\exp(dw)} + \frac{2r^2 w}{3w^2 + w^2 r^{-2}}\right)\hat{k}$

- $C_1\hat{i} + \left(f'(y)\,T(t)y + T(t)\,f(y)\right)\hat{j}$


T
1
Stomata
Transpiration
1
Metabolism
Membrane function
Heating
2
Stomata
6
Stomata
H_2O
Stomata
7
CO_2
Photorespiration
$h\nu$
6
Photoinhibition
inductions by
photoperiod
8
2
3
xylem-
inductions
strategies
Osmotic
consequences
3
Metabolic
Avoidance
9
flow
10
C/N–signalling
Assimilation
11
Sun/shade plants
5
4
Salinity
Ion balances
4
Nutrients
5

Topological Matrix Graph

	1	2	3	4	5	6
1	1	-1	0	0	0	0
2	1	0	0	0	0	-1
3	0	1	0	1	0	0
4	0	0	0	1	1	0
5	0	0	0	0	1	1
6	0	1	0	0	0	-1
7	0	1	1	0	0	0
8	0	0	1	0	0	-1
9	0	0	1	-1	0	0
10	0	1	0	0	1	0
11	0	0	1	0	1	0
12	0	1	1	1	0	0
13	1	0	1	0	0	0

	1	2	3	4	5	6
	-1	-1	0	0	0	0
	0	0	0	0	0	-1
	0	1	0	-1	0	0
	0	0	0	-1	-1	0
	0	0	0	0	-1	-1
	0	1	0	0	0	-1
	0	-1	-1	0	0	0
	0	0	1	0	0	-1
	0	0	1	-1	0	0
	0	-1	0	0	-1	0
	0	0	-1	0	-1	0
	0	-1	-1	-1	0	0
	-1	0	-1	0	0	0

$$
\begin{bmatrix}
S x_1 - x_2 T \\
T x_1 - x_6 T \\
\alpha x_2 + x_4 \alpha \\
| x_4 + x_5 | \\
x_5 + x_6 \\
S x_2 - x_6 S \\
S x_2 + x_3 S \\
\rho x_3 - x_6 \rho \\
x_3 - x_4 \\
x_2 + x_5 \\
CA x_3 + x_5 \frac{C}{N} \\
\cancel{x_2 + x_5 + x_4} \\
H x_1 + x_3
\end{bmatrix}
$$

$$
\begin{bmatrix}
T - x_1 - x_2 S \\
\alpha - x_2 - x_4 \alpha \\
| -x_4 - x_5 | \\
-x_5 - x_6 \\
S - x_2 - x_3 S \\
-x_2 - x_5 \\
\frac{C}{N} - x_3 - x_5 CA \\
\cancel{-x_2 - x_3 - x_4} \\
-x_1 - x_3 H
\end{bmatrix}
$$

Stomata / Transpiration (SIT)

- $\frac{\partial g_s}{\partial g_{s-max}} = C_{gs} \cdot \sqrt{1-\left(\frac{c_i}{c_a}\right)^2} \cdot \frac{w_a}{w_i}$

- $\frac{\partial g_s}{C_{gs}} = g_{s_max} \cdot \sqrt{1-\left(\frac{c_i}{c_a}\right)^2} \cdot \frac{w_a}{w_i}$

- $\frac{\partial g_s}{\partial w_a} = g_{s-max} \cdot C_{gs} \cdot \sqrt{1-\left(\frac{c_i}{c_a}\right)^2} \cdot \frac{1}{w_i}$

- $\frac{\partial g_s}{\partial w_i} = g_{s-max} \cdot C_{gs} \cdot \sqrt{1-\left(\frac{c_i}{c_a}\right)^2} \cdot -\frac{w_a}{w_i^2}$

- $\frac{\partial g_s}{\partial c_a} = g_{s_max} \cdot C_{gs} \cdot \frac{2c_i^2}{c_a^3\sqrt{1-\left(\frac{c_i}{c_a}\right)^2}}$

$$\frac{\partial g_s}{\partial C_i} = g_{s-max} \cdot C_{os} \cdot \frac{2c_i}{2ca^2\sqrt{1-\left(\frac{c_i}{ca}\right)^2}}$$

$$\left(1-\left(\frac{c_i}{c_a}\right)^2\right)^{\frac{1}{2}}$$

$$\frac{1}{2}\left(1-\left(\frac{c_i}{c_a}\right)^2\right)^{-\frac{1}{2}}\left(c_i^2\cdot(c_a)^{-2}\right)$$

$$\left(-2(c_i)^2(c_a)^{-3}\right)$$

$$\left(\frac{-2(c_i)^2}{c_a^3}\right)$$

$$-\frac{2(c_i)^2}{c_a^3\sqrt{1-\left(\frac{c_i}{c_a}\right)^2}}$$

$$\left(1-\left(\frac{c_i}{c_a}\right)^2\right)^{\frac{1}{2}}$$

$$\frac{1}{2}\left(1-\left(\frac{c_i}{c_a}\right)^2\right)^{-\frac{1}{2}}\left(c_i^2\cdot(c_a)^{-2}\right)$$

$$\left(\frac{2c_i}{c_a^2}\right)$$

$$\frac{2c_i}{2c_a^2\sqrt{1-\left(\frac{c_i}{c_a}\right)^2}}$$

Photorespiration (P)

$$\frac{d[C6]Cyto}{dt} = \left(\frac{1}{6} v_{Rubisco_C} + \frac{1}{2} v_{PPDIC} - \frac{1}{2} v_{PEPC} - \frac{1}{27} v_{PEPC} - \frac{1}{6} R_d - \frac{1}{12} v_{Rubisco_O} \right) \cdot \frac{1}{Vol_{Cyto}}$$

$$\frac{V_{Cyto}\, d[C6]Cyto}{dt} = \frac{1}{6} v_{Rubisco_C} + \frac{1}{2} v_{PPDIC} - \frac{29}{54} v_{PEPC} - \frac{1}{6} R_d - \frac{1}{12} v_{Rubisco_O}$$

$$-12\left(\frac{V_{Cyto}\, d[C6]Cyto}{dt} - \frac{1}{6} v_{Rubisco_C} - \frac{1}{2} v_{PPDIC} + \frac{29}{54} v_{PEPC} + \frac{1}{6} R_d \right) = v_{Rubisco_O}$$

Ion Balance (1)

$$V_m = \frac{RT}{F} \ln\left[\frac{P_H [H^+]_o + P_{malat^{2-}} [malate^{2-}]_o + P_K [K^+]_o}{[H^+]_i \quad [malate^{2-}]_i \quad [K^+]_i}\right]$$

C/N Signaling

$$gr = \frac{\ln\left(\frac{PLA(t) \times P_{max}}{\sigma N_{max}(t)}\right)}{\ln\left(\frac{P_{max}}{S(t)}\right) + \frac{L_B(t)}{R_B(t)}}$$

$$gr\left[\ln\left(\frac{P_{max}}{S(t)}\right) + \frac{L_B(t)}{R_B(t)}\right] = \ln\left(\frac{PLA(t) \times P_{max}}{\sigma N_{max}(t)}\right)$$

$$e^{\left(gr\left[\ln\left(\frac{P_{max}}{S(t)}\right) + \frac{L_B(t)}{R_B(t)}\right]\right)} = \frac{PLA(t) \times P_{max}}{\sigma N_{max}(t)}$$

$$\frac{e^{\left(gr\left[\ln\left(\frac{P_{max}}{S(t)}\right) + \frac{L_B(t)}{R_B(t)}\right]\right)^e}}{PLA(t) \times P_{max}} = \sigma N_{max}$$

$$\frac{C_{fix}}{N_{fix}}$$

$f(a,b)$

$$v_{SynPKC} = \frac{v_{max\,SYNPKC} \cdot [Esps,act] \cdot \left(1 + k_{partial} \cdot \frac{[TGP]}{K_{i,TGP}}\right)}{K_{M,ESPS,act} + [SPS],act \left(1 + \frac{[TGP]}{K_{i,TGP}} + \frac{[GlP]}{K_{i,GlP}} + \frac{[GGP]}{K_{i,GlP}} \cdot \frac{[GGP]}{K_{i,GGP}} \cdot \frac{[GGP \cdot TGP]}{K_{i,TGP} \cdot K_{i,GGP}} + \frac{[GGP \cdot TGP]}{K_{i,TGP} \cdot K_{i,GGP}}\right)}$$

a = $v_{SynPKC} \cdot [Esps,act] \cdot \left(1 + k_{partial} \cdot \frac{[TGP]}{K_{i,TGP}}\right)$

b = $K_{M,ESPS,act} + [SPS],act \left(1 + \frac{[TGP]}{K_{i,TGP}} + \frac{[GlP]}{K_{i,GlP}} + \frac{[GGP]}{K_{i,GlP}} \cdot \frac{[GGP]}{K_{i,GGP}} \cdot \frac{[GGP \cdot TGP]}{K_{i,TGP} \cdot K_{i,GGP}} + \frac{[GGP \cdot TGP]}{K_{i,TGP} \cdot K_{i,GGP}}\right)$

$g(c,d)$

$v_{max,SPS} = k_{SPS} \cdot Esps,act = c \cdot d = \frac{a}{b}$

(c = k_{SPS}, d = $Esps,act$)

$k_{SPS} = \frac{v_{max,SPS}}{Esps,total}$ (e = $v_{max,SPS}$, f = $Esps,total$) $h(e,f)$ $\frac{e}{f}$

$v_{SPS} = v_{max,SPS} \cdot \frac{[FGP]}{K_{M,FGP} + [FGP]} \cdot \frac{[UDPG]}{K_{M,UDPG} + [UDPG]}$

(e = $v_{max,SPS}$; p = $\frac{[FGP]}{K_{M,FGP} + [FGP]}$; q = $\frac{[UDPG]}{K_{M,UDPG} + [UDPG]}$) $k(e,p,q)$

$s(r,v)$ $d[Esps,act]/dt = v_{PP} - v_{SynPKC}$ (r = $Esps,act$; v = v_{SynPKC}; v_{PP})

$d[Esps,inact]/dt = v_{SynPKC} - v_{PP}$

(x = v_{SynPKC}; y = v_{PP}) $w(x,y)$

$$f_a = \frac{1}{b}, \quad f_b = -\frac{a}{b^2}$$

$$g_c = d, \quad g_d = c$$

$$h_e = \frac{1}{f}, \quad h_f = -\frac{e}{f^2}$$

$$k_{el} = pq, \quad k_p = k_q = k_{eq} = k_{ep}$$

$$S_r = 1, \quad S_r = -1$$

$$W_x = 1, \quad W_y = -1$$

$$\frac{\frac{b-a}{b^2} + (c+d) + \frac{f-e}{f^2} + (pq+eq+ep)}{\sigma N_{max}}$$

$\frac{C_{fix}}{N_{fix}}$ in variable form

N_{fix}

$f(a,b)$

$$\frac{V_{max\,SNRKI} \cdot [Esps, act] \cdot \left(1 + k_{partial} \cdot \frac{[TGP]}{K_{i,TGP}}\right)}{K_{M,ESPS,\,act} + [SPS],\,act\left(1 + \frac{[TGP]}{K_{i,TGP}} + \frac{[GIP]}{K_{i,GIP}} + \frac{[GLP]}{K_{i,GLP}} \cdot \frac{[GLP]}{K_{i,GLP}} \cdot \frac{[GLP]}{K_{i,TRP}} \cdot \frac{[GLP]}{K_{i,TRP} \cdot K_{i,GLP}} + \frac{[GLP]^2}{K_{i,TRP} \cdot K_{i,GLP}}\right)}$$

$g(c,d)$

a

b

$\frac{C_{fix}}{N_{fix}}$ with the values plugged in $\left(\frac{C}{N}\right)$

$$\left[\frac{\left(K_{M,ESPS,\,act} + [SPS],\,act\left(1 + \frac{[TGP]}{K_{i,TGP}} + \frac{[GIP]}{K_{i,GIP}} + \frac{[CCK]}{K_{i,GIP}} + \frac{[GLP]}{K_{i,GLP}} \cdot \frac{[CCK,TRP]}{K_{i,TRP}} \cdot \frac{[GLP,TRP]}{K_{i,TRP} \cdot K_{i,GLP}}\right)\right) - \left(V_{max\,SNRKI} \cdot [ESPS,act] \cdot \left(1 + k_{partial} \cdot \frac{[TGP]}{K_{i,TGP}}\right)\right)}{\left(K_{M,ESPS,\,act} + [SPS],\,act\left(1 + \frac{[TGP]}{K_{i,TGP}} + \frac{[GIP]}{K_{i,GIP}} + \frac{[CCK]}{K_{i,GIP}} + \frac{[GLP]}{K_{i,GLP}} \cdot \frac{[CCK,TRP]}{K_{i,TRP}} \cdot \frac{[GLP,TRP]}{K_{i,TRP} \cdot K_{i,GLP}}\right)\right)^2}\right]$$

$$+ \left(K_{SPS} + Esps,\,act\right) + \left[\frac{Esps,\,Total - V_{max,\,SPS}}{(Esps,\,Total)^2}\right] - \left[\left(\frac{[FGF]}{K_{M,FGF} + [FGF]}\right)\left(\frac{[UDPG]}{K_{M,UDPG} + [UDPG]}\right)\right.$$

$$\left. + \left(V_{max,\,SPS}\right)\left(\frac{[UDPG]}{K_{M,UDPG} + [UDPG]}\right) + \left(V_{max,\,SPS}\right)\left(\frac{[FGF]}{K_{M,FGF} + [FGF]}\right)\right]$$

$$e^{\left(gn\left[\ln\left(\frac{P_{max}}{s(t)}\right) + \frac{L_0(t)}{R_0(t)}\right]\right)}$$

$$\overline{\rho LA(t) \times P_{max}}$$

Osmotic Consequences (OC)

$$\dot{y} = +u_1 = c_1 x - C_2 f(y) T(t) y$$

$$\dot{x}: -u_1 + u_2 = -c_1 x + c_2 f(y) T(t) y + \frac{c_3 w}{x} - c_4 x$$

$$\frac{\partial \dot{y}}{\partial x} = c_1$$

$$\frac{\partial \dot{y}}{\partial y} = f'(y) T(t) y + f(y) T(t)$$

$$\boxed{c_1 - C_2 f'(y) T(t) y - f(y) T(t)}$$

$$\frac{\partial \dot{x}}{\partial x}: -c_1 - \frac{C_3 w}{x^2} - c_4$$

$$\frac{\partial \dot{x}}{\partial y} = C_2 f'(y) T(t) y - f(y) T(t)$$

$$\boxed{-\frac{C_3 w}{x^2} - C_2 f'(y) T(t) y - f(y) T(t) - c_1 - c_4}$$

$$C_1 - C_2 f'(y) T(t) y - f(y) T(t)$$

$$-\frac{C_3 w}{x^2} - C_2 f'(y) T(t) y - f(y) T(t) - C_1 - C_4$$

$$-\frac{C_3 w}{x^2} - 2 C_2 f'(y) T(t) y - f(y) T(t) - C_4$$

Heating (T)

$$f_M(z, M, T_i) = \exp\left[-\left(\frac{z}{\mu}\right)^{c_3}\right] \frac{M_{max}\left[\frac{T_H - T_i}{T_H - T_L}(1-\alpha_2)+\alpha_2\right] - M}{\alpha_2 M_{max}\left[\frac{T_H - T_i}{T_H - T_L}(1-\alpha_2)+\alpha_2\right] - \left(\alpha_2 M_{max}\left[\frac{T_H - T_i}{T_H - T_L}(1-\alpha_2)+\alpha_2\right] - M\right)}$$

$$\left(A_{SF,max}(T_i)\right)\left(\frac{dv}{dt}(T_i)\right) f_{\psi_i}(\psi_i) \times \exp\left[-\left(\frac{z}{\mu}\right)^{c_3}\right] \frac{M_{max}\left[\frac{T_H - T_i}{T_H - T_L}(1-\alpha_2)+\alpha_2\right] - M}{\alpha_2 M_{max}\left[\frac{T_H - T_i}{T_H - T_L}(1-\alpha_2)+\alpha_2\right] - \left(\alpha_2 M_{max}\left[\frac{T_H - T_i}{T_H - T_L}(1-\alpha_2)+\alpha_2\right] - M\right)}$$

$A_{SF,max}(T_i)$ → $a\hat{i}$

$$A_c(C_x, T_i) = \left[\frac{V_{c,max0}\exp\left[\frac{H_{a1}}{RT_i}\left(1-\frac{T_i}{T_1}\right)\right]}{1+\exp\left(\frac{S_{rc}T_i - H_{d1}}{RT_i}\right)}\right] \frac{C_x - \left[\Gamma_0 + \Gamma_1(T_i - T_0) + F_2(T_i - T_0)^2\right]}{C_x + \left[K_{c0}\exp\left[\frac{H_{Cx}}{RT_0}\left(1-\frac{T_i}{T_i}\right)\right]\right]\left(1+\theta_i/\left[k_{0,exp}\left[\frac{H_{ex}}{RT_0}\left(1-\frac{T_0}{T_i}\right)\right]\right]\right.}$$

$b\hat{j}$ → $d\hat{i}$

$$J_{max}(T_i) = J_{max0}\left\{\frac{\exp\left[\frac{H_{a1}}{RT_0}\left(1-\frac{T_0}{T_i}\right)\right]}{1+\exp\left(\frac{S_{rQ}T_i - H_{dJ}}{RT}\right)}\right\}$$

S

r

p

z

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & 0 \\ d & 0 & 0 \end{vmatrix}$$

$$(0-0)\hat{i} - (0-0)\hat{j} + (0-bd)\hat{k}$$

$$= -(bd)\hat{k}$$

$$0\hat{i} + 0\hat{j}$$

$$-\left[\left(\text{[illegible]}\,\rho_{air}(T_d)\,x\,f_{\psi_i}(\psi_1)\times \exp\left[-\left(\frac{z}{\Delta z}\right)^{C_2}\right]\frac{M_{max}\left[\frac{T_{in}-T_l}{T_{in}-T_c}(1-\alpha_s)+\alpha_s\right]-M}{\alpha_s M_{max}\left[\frac{T_{in}-T_l}{T_{in}-T_c}\right]-\left(\alpha_s M_{max}\left[\frac{T_{in}-T_l}{T_{in}-T_c}(1-\omega)+\omega\right]-H\right)}\right)\cdot\left(\frac{\left[V_{c,max}\exp\left[\frac{H_a}{RT_l}\left(1-\frac{T_l}{T}\right)\right]\right]\left[c_p-\left[\tau_l+\tau_r(T-T_r)\right]F_c(T-T_r)\right]}{1+\exp\left(\frac{S\sigma\varepsilon T-H_d}{RT_l}\right)}\right)\right]\hat{k}$$

$$\sqrt{\qquad\qquad\qquad

$$\frac{S}{r} + \frac{p}{r} + p\left(-\frac{S}{r^2}\right)$$

$$\frac{\exp\left[\frac{H_{dJ}}{RT_0}\left(1-\frac{T_0}{T_1}\right)\right]}{1+\exp\left(\frac{S_{rQ}T_1 - H_{dJ}}{RT}\right)}\hat{i} + \frac{J_{max0}}{1+\exp\left(\frac{S_{rQ}T_1 - H_{dJ}}{RT}\right)}\hat{j} + J_{max0}\left(-\frac{\exp\left[\frac{H_{dJ}}{RT_0}\left(1-\frac{T_0}{T_1}\right)\right]}{1+\exp\left(\frac{S_{rQ}T_1 - H_{dJ}}{RT}\right)}\right)\hat{k}$$

Gradient

$$D_{Temp}$$

$$-\left(\frac{-\left[\frac{H_{dJ}}{RT_0}\left(1-\frac{T_0}{T_1}\right)\right]}{1+\exp\left(\frac{S_{rQ}T_1 - H_{dJ}}{RT}\right)}\right)\cdot\left(\frac{\left[V_{c,max}\frac{\exp\left[\frac{H_{dV}}{RT_0}\left(1-\frac{T_0}{T_1}\right)\right]}{1+\exp\left(\frac{S_{rQ}T_1 - H_{dJ}}{RT}\right)}\right]}{\frac{\exp\left(\frac{H_{dV}}{RT_0}\right)}{K_c}}\right)\left(-\frac{\exp\left[\frac{H_{dJ}}{RT_0}\left(1-\frac{T_0}{T_1}\right)\right]}{1+\exp\left(\frac{S_{rQ}T_1 - H_{dJ}}{RT}\right)}\right)$$

$$\sqrt{-\left(\frac{-\left[\frac{H_{dJ}}{RT_0}\left(1-\frac{T_0}{T_1}\right)\right]}{1+\exp\left(\frac{S_{rQ}T_1 - H_{dJ}}{RT}\right)}\right)\cdot\left(\frac{\left[V_{c,max}\frac{\exp\left[\frac{H_{dV}}{RT_0}\left(1-\frac{T_0}{T_1}\right)\right]}{1+\exp\left(\frac{S_{rQ}T_1 - H_{dJ}}{RT}\right)}\right]}{\frac{\exp\left(\frac{H_{dV}}{RT_0}\right)}{K_c}}\right)}^2$$

CAM Assimilation (CA)

$$-\left(\frac{C_{ext}(t) - w}{\exp(\alpha w)} - L(t)w + C_7 L_{ck} \frac{w_r}{L_{st} + L_{ck}} \frac{w_r}{w + w_r}\right)$$

$$2\left(\frac{C_{ext}(t) - w}{\exp(\alpha} - C_6 L(t) w + C_7 L_{ck} \frac{w_r}{L_{st} + L_{ck}} \frac{w_r}{w + w_r}\right)$$

↓

$$\frac{C_{ext}(t) - w}{\exp(\alpha w)} - C_6 L(t) w + C_7 L_{ck} \frac{w_r}{L_{st} + L_{ck}} \frac{w_r}{w + w_r}$$

$$\frac{\partial u_3}{\partial w} = \frac{-1}{\exp(\alpha w)} + \frac{\alpha w}{\exp(\alpha w)} - C_6 L L(t) + \frac{C_7 L_{ck}}{L_{st} + L_{ck}} \left(\frac{-w_r - w_r^2}{(w + w_r)^2}\right)$$

$$\frac{\partial u_3}{\partial w_r} = \frac{C_7 L_{ck}}{L_{st} + L_{ck}} \frac{w - w^2}{(w + w_r)^2} = -\frac{C_7 L_{ck} w - C_7 L_{ck} w^2}{L_{st} + L_{ck} (w + w_r)^2}$$

$$\frac{\partial u_3}{\partial t} = \frac{C_{ext}'(t)}{\exp(\alpha w)} - C_6 L w L'(t)$$

$$\frac{\partial u_3}{\partial L_{lk}} = C_7 \left(\frac{w_r}{w + w_r} \right)$$

$$\frac{\partial u_3}{\partial L_{lt}} = -C_7 \frac{L_{lk}}{L_{lt}^2} \left(\frac{w_r}{w + w_r} \right)$$

Full Slope

$$\frac{-1}{\exp(\alpha w)} + \frac{\alpha w}{\exp(\alpha w)} - C_6 L L(t) + \frac{C_7 L_{lk}}{L_{lt} + L_{lt}} \left(\frac{-w_r - w_r^2}{(w + w_r)^2} \right) - \frac{C_7 L_{lk} w - C_7 L_{lk} w^2}{L_{lt} + L_{lk} (w + w_r)^2} + \frac{C_{ext}'(t)}{\exp(\alpha w)} - C_6 L w L'(t)$$

$$+ C_7 \left(\frac{w_r}{w + w_r} \right) - \frac{C_7 L_{lk}}{L_{lt}^2} \left(\frac{w_r}{w + w_r} \right)$$

Membrane Function (M)

- hours per cycle
- Sum up all the integrals because the additional accumulation of Carbon may vary

$$\sum_{n=1}^{12} \int \left(\frac{-1}{u\exp(\alpha w)} + \frac{\alpha w}{u\exp(\alpha w)} - C_4 L\, L(t) + \frac{C_2 L_b}{L_d + L_b}\left(\frac{-W_T - W_T^2}{(W_T + W_T)^2}\right) - \frac{-C_7 L_b W_T - C_7 L_b W_T^2}{L_d + L_b (W_T + W_T)^2} + \frac{C_{ens}'(t)}{u\exp(\alpha w)} - C_6 L\, w L'(t) + C_7 \left(\frac{W_T}{W_T + W_T}\right) - \frac{C_7 L_b}{L_d^2}\left(\frac{W_T}{W_T + W_T}\right) \right)(n) \; \frac{RT}{F} \ln\left[\frac{p_H [H^+]_0 + p_{malate^{2-}} [malate^{2-}]_0 + p_K [Cl^-]_0}{[H^+]_i \quad [malate^{2-}]_i \quad [Cl^-]_i} \right] d\,Vm$$

$$\left(\frac{-1}{u\exp(\alpha w)} + \frac{\alpha w}{u\exp(\alpha w)} - C_4 L\, L(t) + \frac{C_2 L_b}{L_d + L_b}\left(\frac{-W_T - W_T^2}{(W_T + W_T)^2}\right) - \frac{-C_7 L_b W_T - C_7 L_b W_T^2}{L_d + L_b (W_T + W_T)^2} + \frac{C_{ens}'(t)}{u\exp(\alpha w)} - C_6 L\, w L'(t) + C_7 \left(\frac{W_T}{W_T + W_T}\right) - \frac{C_7 L_b}{L_d^2}\left(\frac{W_T}{W_T + W_T}\right) \right)(n-1)$$

Night Influx

$$\sum_{n=1}^{12} \int \left(\frac{-1}{u\exp(\alpha w)} + \frac{\alpha w}{u\exp(\alpha w)} - C_4 L\, L(t) + \frac{C_2 L_b}{L_d + L_b}\left(\frac{-W_T - W_T^2}{(W_T + W_T)^2}\right) - \frac{-C_7 L_b W_T - C_7 L_b W_T^2}{L_d + L_b (W_T + W_T)^2} + \frac{C_{ens}'(t)}{u\exp(\alpha w)} - C_6 L\, w L'(t) + C_7 \left(\frac{W_T}{W_T + W_T}\right) - \frac{C_7 L_b}{L_d^2}\left(\frac{W_T}{W_T + W_T}\right) \right)(n-1) \; \frac{RT}{F} \ln\left[\frac{p_H [H^+]_0 + p_{malate^{2-}} [malate^{2-}]_0 + p_K [Cl^-]_0}{[H^+]_i \quad [malate^{2-}]_i \quad [Cl^-]_i} \right] d\,Vm$$

$$\left(\frac{-1}{u\exp(\alpha w)} + \frac{\alpha w}{u\exp(\alpha w)} - C_4 L\, L(t) + \frac{C_2 L_b}{L_d + L_b}\left(\frac{-W_T - W_T^2}{(W_T + W_T)^2}\right) - \frac{-C_7 L_b W_T - C_7 L_b W_T^2}{L_d + L_b (W_T + W_T)^2} + \frac{C_{ens}'(t)}{u\exp(\alpha w)} - C_6 L\, w L'(t) + C_7 \left(\frac{W_T}{W_T + W_T}\right) - \frac{C_7 L_b}{L_d^2}\left(\frac{W_T}{W_T + W_T}\right) \right)(n)$$

Day Influx